



UNIVERSIDAD TECNICA FEDERICO SANTA MARIA

Master's Thesis

Design and Sizing of an Energy Storage System for a Hybrid Tugboat

Thesis as a requirement to obtain the degree of
Magister en Ciencias de la Ingeniería Electrónica

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Abstract

The global maritime industry is increasingly prioritizing sustainability, promoting the adoption of various hybrid and fully electric propulsion systems for different vessel types. This document presents a design and sizing methodology of an energy storage system for a hybrid tugboat, customized to meet the operational demands while minimizing environmental impact. The proposed framework integrates analysis of load profile, energy storage sizing, battery technologies selection, and propulsion system optimization to achieve an optimal balance between performance, and emissions reduction. The results demonstrate that a well-designed energy storage system for a hybrid tugboat can achieve reductions in fuel consumption and greenhouse gas emissions without compromising the availability and robustness required for towing and transit operations.

Resumen

La industria marítima mundial está dando cada vez más prioridad a la sostenibilidad, promoviendo la adopción de diversos sistemas de propulsión híbridos y totalmente eléctricos para diferentes tipos de buques. En este documento se presenta una metodología de diseño y dimensionamiento de un sistema de almacenamiento de energía para un remolcador híbrido, personalizado para satisfacer las exigencias operativas y minimizar el impacto medioambiental. El marco propuesto integra el análisis del perfil de carga, el dimensionamiento del almacenamiento de energía, la selección de tecnologías de baterías y la optimización del sistema de propulsión para lograr un equilibrio óptimo entre el rendimiento y la reducción de emisiones. Los resultados demuestran que un sistema de almacenamiento de energía bien diseñado para un remolcador híbrido puede lograr reducciones en el consumo de combustible y las emisiones de gases de efecto invernadero sin comprometer la disponibilidad y la robustez necesarias para las operaciones de remolque y tránsito.

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Chapter 1

Introduction

1.1 Motivation and Background

The growing global concern over climate change has significantly pressured industries to adopt sustainable practices. The maritime sector, responsible for approximately 2.5% of energy sector CO₂ emissions as shown in figure 1.1, has continuously increased in emissions [1] [2].

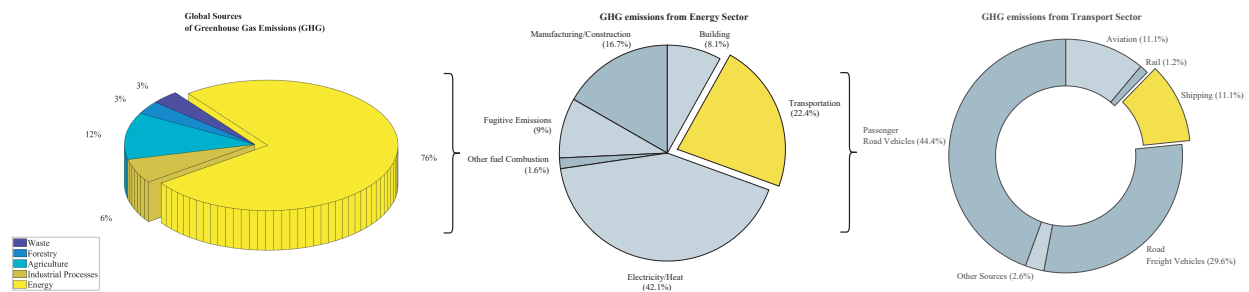


Figure 1.1: Global sources of greenhouse gas emissions and breakdown of CO₂ emissions from the energy sector and the transport subsector [1].

The International Maritime Organization (IMO) is implementing strategies to reduce the emissions from international shipping by establishing standards, imposing restrictions and providing incentives. The initial strategy avoids a reduction in total GHG emissions from international shipping by at least 50% by 2030 compared to 2008, consistent with the Paris Agreement. IMO's current global standard for SO_x emissions is 0.5% mass by mass (m/m) by 2030 per the **International Convention for the Prevention of Pollution by Ships (MARPOL 73/78) Annex VI**, as shown in figure 1.2a.

Similarly, the emission limit for NO_x from engines is dependent on the type of the engine and the nominal engine speed, as shown in figure 1.2b. The global limit has been set to 14.4 g/kWh for engines with speeds less than 130 rev/min and 7.7 g/kWh for engines with speed greater than 2000 rev/min.

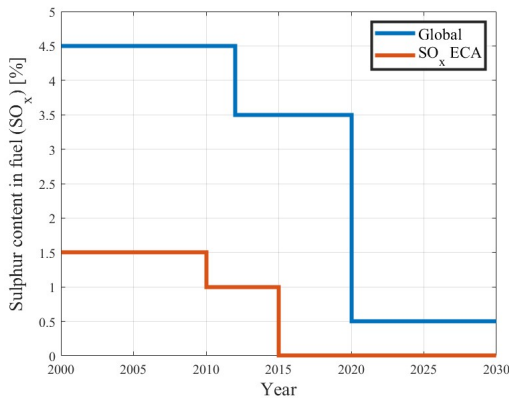
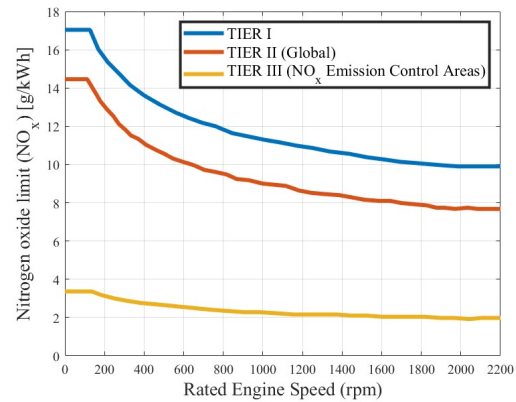
(a) Changes in SO_x limits over time.(b) NO_x limits in terms of engine speed.

Figure 1.2: The MARPOL Annex VI emission limits for global and Emission Controlled Areas (ECAs) perspective.

With the **Net-Zero Framework (NZF)** approved by the IMO in April 2025, the course is being set for the global maritime fuel transition [3]. The NZF aims to meet the emission reduction targets set out in the 2023 IMO GHG Strategy. Pending adoption in October 2025, a GHG fuel intensity (GFI) metric, in combination with a two-tier GHG pricing mechanism, will require operators to reduce their ship's GHG emissions intensity by 21% by 2030, with financial penalties for those who fail to do so. Beyond 2030, the requirements become rapidly stricter.

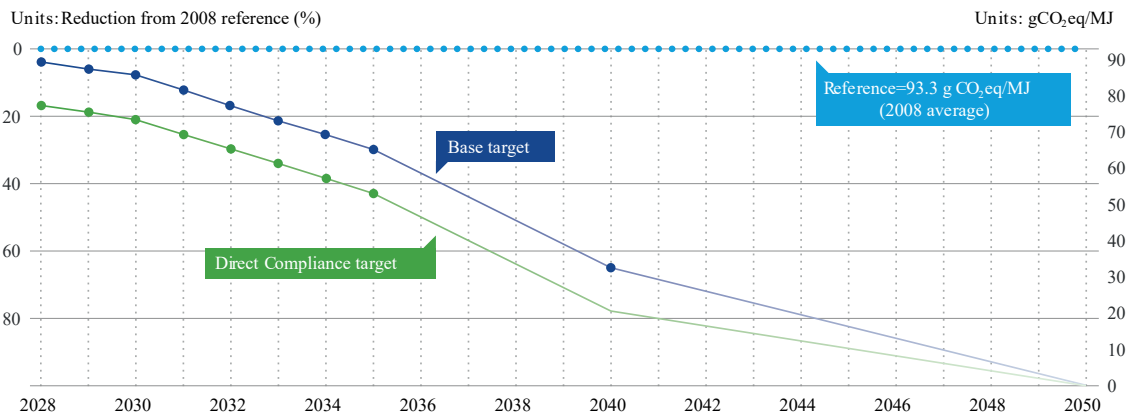


Figure 1.3: GFI reduction factors and reference value in NZF [3].

The NZF promotes low-GHG fuels and technologies to comply with the IMO's 2023 Strategy: reducing emissions by 20% by 2030 and 70% by 2040 (compared to 2008), reaching net zero by 2050, as is shown in fig 1.3. The NZF regulates the GFI metric (gCO₂eq/MJ) under a well-to-wake approach, tightening limits from 2028 onwards. This system of incentives and penalties requires shipowners to evaluate solutions that minimize consumption and reduce the carbon intensity of their vessels.

Many vessels chartered in the coming years could still be in operation in 2050, so it is necessary to take into account the strict requirements that lie ahead in order to maintain their commercial appeal, asset value, and profitability in the coming decades. Furthermore, beyond 2030, ships in operation may need to consider modernization options that allow for the use of fuels with low

greenhouse gas emissions.

The incentives to reduce emissions and improve the efficiency of equipment are motivating ship manufacturers to adopt zero-emission technologies. Fuel cells and energy storages (for example, batteries and supercapacitors) reduce GHG emissions and are the key enablers for the realization of low-emissions operation of marine vessels.

1.1.1 Propulsion Architectures

While conventional maritime transport is characterized by restricting the amount of fuel produced by diesel engines in propulsion systems through Tier standards, which translates into a continuous decrease in fossil fuel consumption, fully electric and hybrid propulsion architectures, made possible by innovative high-performance energy storage systems (batteries, fuel cells, supercapacitors, etc.), are a promising way to significantly reduce fossil fuel consumption, pollutants, and greenhouse gas emissions [5].

New developments of electrically propelled ships are based on an integrated energy distribution system architecture that releases large propulsion machines [6].

For full-electric propulsion, the main advantages include better dynamic performance (start-up, arrest, changes in speed, and regenerative braking), reduced fuel consumption due to the more efficient and flexible use of generators that work within their minimum specific fuel consumption region, and the possibility of having shorter shaft lines because the electric motor must be installed close to the propeller.

The hybrid propulsion system uses a combination of diesel engines or gas turbines with electric motors to provide a wide range of propeller speeds [6]. One of the main advantages of this configuration concerns an increase in efficiency at low loads. Hence, less fuel consumption, fewer emissions, and a small footprint are achieved. Hybrid propulsion systems (HPS) are interesting when there is a large variation in power demand during operations. The design rules of this type of systems can differ significantly for different types of vessel, mainly due to the different types of mission profiles and operation requirements [5].

Hybridization of the propulsion chains are based on the separation or simultaneous use of several different energy sources. The choice of different power sources, and sources combinations is designed to adapt to the ship power and energy requirements which include the needs of low and high power, the needs of flexibility in changes of rhythm and the needs of redundancy and reliability.

As can be seen in the following subsections, there are three main architectures for hybrid propulsion chains [5]:

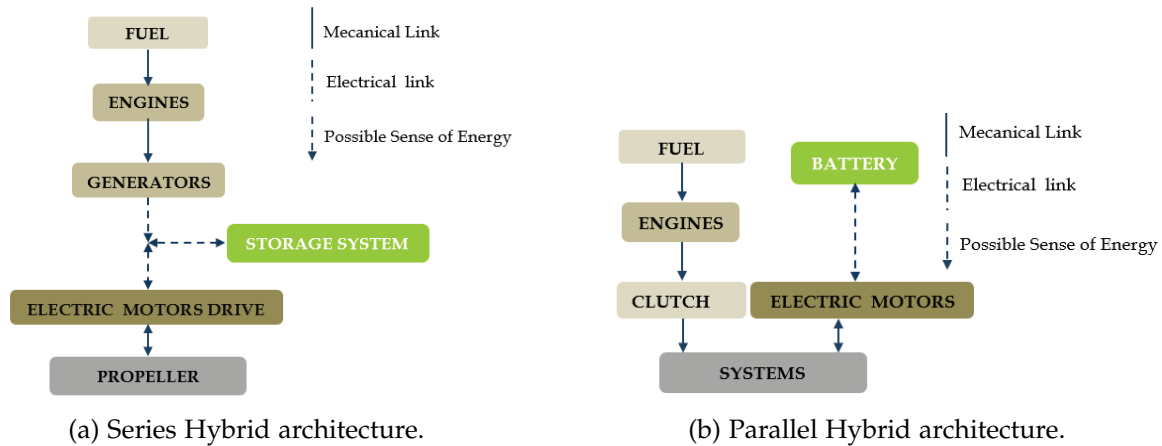


Figure 1.4: Hybridization schemes [5].

a) Series Hybrid Architectures

In this type of HPS architecture, two kind of energy system (combustion engines and electric systems) are used in series. The propulsion is done by a variable speed electrical drive (propulsion motor and power converter), as can be seen in the figure 1.4a. The electrical power of the propulsion motor is generally supplied by a set of generators driven by combustion engines. Another option is to use fuel cells to provide electric power or a combination of fuel cells and generators driven by combustion engine. Energy storage systems (ESS) (battery, supercapacitors, etc.) can be connected to the electrical bus to provide propulsion power. This ESS can be used separately or with the generators. In this configuration, electric power is used as an energy vector between the combustion engine and the propulsion motor, so there will be no direct mechanical connection between the engine and the propeller. This configuration allows dissociating the operating points of the combustion engine and of the propeller. This is why it allows increasing the global efficiency of the system by optimizing the operating point of combustion engine and propeller in efficiency point of view. Of course, this advantage would be more significant if several generators and ESS are used, because it allows providing the required propulsion power by choosing an optimal combination of sources in order to keep the running combustion engines in their optimal operating area. However, for small ships, the possibility of multiplying the sources is very limited by volume and mass constraints.

b) Parallel Hybrid Architectures

Parallel hybrid architecture combines two kinds of propulsion motors: electric motors and combustion engines which are mechanically coupled by the same shafts through gearboxes and clutches, as can be seen in the figure 1.4b. The electrical motors and drive are supplied by electrical ESS and/or independent power sources generators driven by combustion engines and/or Fuel Cell for example. The two propulsion systems can be used together or separately. In most cases one of the propulsion systems is devoted to be used for low speed operations (classically the electric system) and the other one (classically the combustion engine) is used for high speeds. The first system can also be used as a boost system to provide a supplementary power during transient (starting, acceleration, etc.). This architecture reduces the number of elements compared to hybridization series and allows optimizing the sizing of each of the energy sources.

c) Series-Parallel Hybrid Architectures

The series-parallel architecture combines both architectures defined previously. This association allows the passage of a parallel-type operation to a serial type operation.

Hybrid Propulsion systems are based on the combination of several power sources and energy storage systems.

1.1.2 Power Generation Systems

There are different power generation systems associated with propulsion systems for boats. The main ones are listed below:

a) Steam Turbine

A steam turbine is a device that extracts thermal energy from pressurized steam and uses it to do mechanical work on a rotating output shaft [11] [12]. Steam turbines work by expanding steam through a series of blades mounted on a shaft, which turns a propeller to move the ship. They are particularly favoured for their smooth operation, high power output, and efficiency at high speeds, making them suitable for large vessels like liquefied natural gas (LNG) carriers and naval ships.

Steam turbines revolutionized ship propulsion with their introduction in the late 19th century, offering advantages such as small size, reduced maintenance, and low vibration. Despite the need for precise reduction gears or turbo-electric transmission to match the high RPM of turbines to the effective propeller speeds, the benefits included light weight and compact engine rooms.

b) Gas Turbine

Gas turbine engines are defined as aerospace engines that convert the chemical energy in fuel into thermal energy through combustion with air, resulting in the rapid expansion of gas that can perform work in the turbine and exhaust nozzle [13].

The gas turbine engine is recognized for its relative mechanical simplicity and its capacity to use a broad spectrum of fuels. The excellent power-weight ratio of gas turbine engines can be effectively exploited. The performance of gas turbine engines is affected by compressor efficiency, turbine efficiency, pressure ratio, peak cycle temperature, and combustion efficiency. Compressor and turbine efficiencies are continuously increasing through better design and manufacturing techniques, but little room is left for improvement.

The first time a gas turbine was mounted on a watercraft occurred in 1947 in the United Kingdom, when a diesel free power turbine was mounted on a modified gunboat as a proof of concept [14]. This craft endured sea trials for 4 years, until the technology was proven to be a viable concept for the future of naval propulsion. The HMS Grey Goose became the first ship powered entirely by gas turbines, launched in the United Kingdom in 1953, fulfilling its mission to demonstrate the viability of gas turbine propulsion at sea. Since the inception of this technology, gas turbine engines or combinations of them with other engine technologies have been utilized in a wide range of watercraft, ranging from hovercraft all the way up to full-size aircraft carriers, with power outputs starting at single digits of MW all the way to hundreds of MW.

c) Diesel Engines

A diesel engine is a prime mover with the function to convert chemical energy stored in fuel to mechanical energy [5] [7]. Compared to other prime movers such as a gas turbine or steam

turbine, the diesel engine has a high efficiency. Its low fuel consumption is the main reason it is widely used in marine applications.

This power source is characterized by a high level of autonomy and by a relatively low cost compared to other sources. Diesel engines are characterized by a high level of greenhouse gases and pollutants emission particularly if they are used at a lower level of power than their rated power.

d) Fuel Cells

Fuel Cells (FCs) work as power sources without greenhouse gas emissions [29] [29]. Instead of burning fuels, FCs have various techniques to transfer chemical energy to electric energy, which include:

- Alkaline Fuel Cell (AFC)
- Proton Exchange Membrane FC (PEMFC)
- Direct Methanol FC
- Solid Oxide FC (SOFC)

The fuel in FC can be methanol, diesel, hydrogen, LNG, and so on. The dc output of FCs is converted to the required voltage through dc/dc converters to fit the dc bus.

FCs have the advantage of a long lifetime, which is attractive in marine applications. However, the slow dynamic response is the main barrier of using FCs in shipboard microgrids due to their slow internal electrochemical and thermodynamic responses [29].

1.1.3 Energy Storage Systems

An intermediate energy storage system can be used to better manage energy and reduce the size of power generation system. The integration of energy storage systems offers interesting solutions for energy efficiency and optimal energy use.

a) Ultra-capacitors

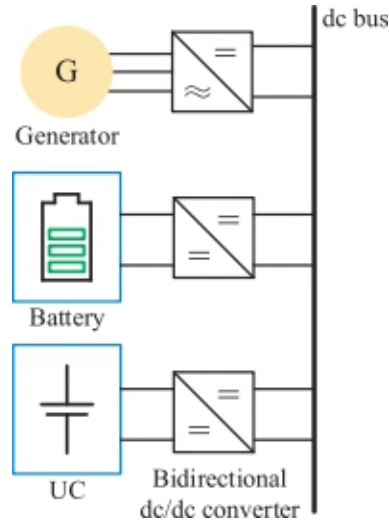


Figure 1.5: Block diagram of dc bus with UC-ESS [29].

Ultra-Capacitors Energy Storage Systems (UCSS), also known like Supercapacitors, can be interesting intermediate energy storage systems with short power transients [5]. They are characterized by a high value of charge and discharge peak power and can be used to rapid dynamic storage with very low specific energy. They are usually associated with other systems, such as batteries, generators and fuel cells, to manage high power short transients, as is shown in figure 1.5. Consequently, combining UCs and batteries not only can meet the long-term load demand but also reduce peak power stresses on the battery packs [29].

b) Flywheels

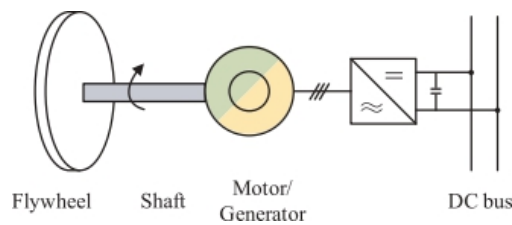


Figure 1.6: Structure of the flywheel ESS (FESS) [29].

Flywheels Energy Storage Systems (FESS) can be used in some particular cases. Like ultra-capacitors, they are characterized by a high power density but a relatively low energy density. Currently, FESS is becoming widely applied in various applications including the marine industry.

Typically, FESS is a kind of electric supply device that stores energy in the form of kinetic energy in a rotating flywheel connected to the shaft of an electric machine [29]. The FESS structure is depicted in Fig. 1.6, containing rotor (flywheel), shaft, motor/generator, and power electronic interface. The amount of stored energy depends on the form, mass, and rotation speed of the flywheel. In charging mode, the flywheel speeds up and store the kinetic energy in the high-speed

rotational disk; while in discharging mode, the stored kinetic energy will be released by slowing down the rotating flywheel.

However using flywheels leads to implement safety systems to protect crew and passengers from possible flywheel failure. So their use in small ship context seems very limited.

c) Batteries

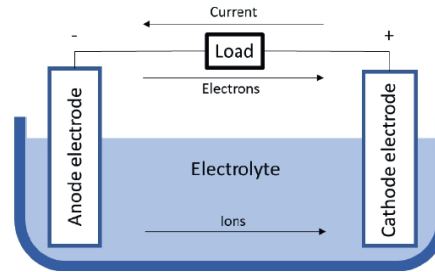


Figure 1.7: Components of a battery [19].

Table 1.1: Energy Storage Systems Comparison [21].

Technology	Battery	Supercapacitor	Flywheel
Energy Density (Wh/kg)	50-250	0.05-5	5-100
Power Density (W/Kg)	50-2000	100k	1000
Lifecycle (cycle)	400-9000	50k	20k-100k
Lifetime (years)	5-10	5-8	15-20
Overall Efficiency (%)	85-99	60-65	15-20

The oldest and most classical method for storing electrical energy is using batteries. In general, as can be seen in fig. 1.7, a battery is comprised of two different poles – a positive electrode called the cathode and a negative electrode called the anode. Then some material is used inside the battery, called electrolyte, that enables ions, or charge carriers, to be transferred back and forth between these poles by electrochemical reactions [19]. A separator can be placed between the cathode and anode, preventing them from touching each other. Hence, when the poles are connected by an electrical conducting material, electrons will flow through the external electrical circuit, and ions to flow through the electrolyte. This process allows energy to be stored or produced in the battery. The chosen energy carrier material, electrode and electrolyte composition, and the shape of the electrodes determine the properties of the batteries.

Battery technology has matured significantly during the past two decades as not only the world but also energy has become increasingly mobile [20].

As shown in the table 1.1, the battery provides the highest power density but lacks overall efficiency compared to energy storage systems. The flywheel, on the other hand, is the most durable system, offering a cycle life of between 20,000 and 100,000 cycles and a service life of between 15 and 20 years. However, due to its low power density, it may not provide enough power for a vessel.

The comparison seen in table 1.1 helps to understand why batteries are usually chosen over other types of energy storage systems: because of their reliability and high energy density.

1.1.4 Battery Systems in Maritime Applications

Battery application onboard ships can have multiple functional roles. While batteries can fully power a vessel for short distance or duration, improving performance and energy efficiency of the overall vessel is often the key purpose.

In 2018, the European Maritime Safety Agency (EMSA) commissioned DNV GL to perform a study on the use of electrical storage systems in shipping, with the objective of providing an overview of technology, research, feasibility, regulations and safety of battery systems in maritime applications [19]. Figure 1.8 shows the different uses of batteries recognized by EMSA:

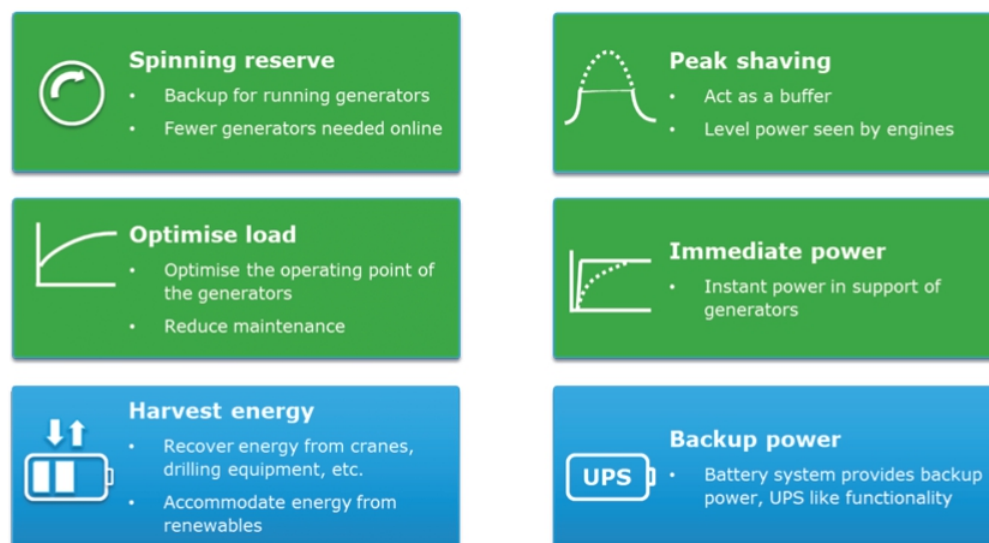


Figure 1.8: Functional roles of battery systems onboard ships [19].

- **Spinning reserve:** This mainly applies to diesel electric dynamic positioning vessels. Using a battery-provided spinning reserve allows the vessel to reduce the number of generators running required to maintain the spinning reserve. It allows for lower fuel consumption, lower emissions, higher efficiency of running engines and lower maintenance costs due to reduced running hours [23].
- **Peak shaving:** Battery energy storage systems also provide additional electric power required for short-term additional power demands. In conventional propulsion systems, it is necessary to run additional diesel generator to meet those demands. Using a battery instead, the additional generator can be turned off, therefore saving on fuel and wear and tear of the engine. Vessels equipped with heavy electricity consumers such as cranes or electric-driven thrusters will benefit from this arrangement [23].
- **Optimise load:** This concept is recognized also like *Load Leveling*. BESSs are used to keep the load of the diesel engine/generator constant, allowing them to operate at maximum efficiency by charging when the load demand is low, then contributing to the fulfilment of the load request [25].
- **Immediate power:** This concept is recognized also like *Ramp Levelling* and is used to enhance dynamic engine performance [25]. Gas engines and other energy sources are not capable of handling fast load variations. In such cases, a BESS can be utilized to take care of the variations, while the main energy source produces slowly varying power [23].

- **Harvest energy:** This service consists of a time shift of energy generation by storing energy from photovoltaic panels, and/or other renewable sources such as wind, and then releasing it when they are absent. In addition, regenerative power improves ship energy management, using BESS to store energy obtained from the regenerative process of double speed engine braking (e.g., ships with cranes or large winches) [25].
- **Backup power:** BESS allows the vessel to stop diesel engines when onboard electricity consumption is low (e.g., in port). The generators are started only to charge the batteries. It is possible to reduce emissions because the diesel engines operate at optimal load (with maximum efficiency and minimum specific fuel consumption) during charging [23].

Additionally, other authors acknowledge the service of the **zero-emission mode** [25], that utilizes full electric propulsion for a limited time, reducing ship noise and vibration. Here, a BESS allows for switching-off all the onboard internal combustion engines. However, when fully powered by its own BESS, these ships are subject to specific technical limitations, such as max speed and maximum operation time, depending on battery power and energy content.

Among the commercially available battery technologies that can be used to provide services to boats, considering that is not feasible or possible to cover the entire spectrum of all technologies, because for any given chemistry there is most often a wide range of products representing different levels of quality and performance, information on some of them is provided [19]:

- **Lead-acid:** Lead-acid batteries are supplied worldwide by a large supplier base at a very low cost. Automotive batteries and industries where standby electrical power is critical are the biggest markets. It is considered to be very safe, since the electrolyte and active materials are not flammable; although the batteries are known to produce hydrogen under charging. The main drawbacks for such batteries are the specific energy and energy density. The power density is considered high for lead acid battery but is also here outperformed by Li-ion.
- **Nickel Cadmium (Ni-Cd):** Nickel cadmium batteries are mature technologies, that are widely used in UPS applications, for example. This technology is economically priced and presents the lowest per cycle cost. Good ventilation is important in the room to avoid explosive concentration of hydrogen. Ni-Cd have memory effect that causes a loss of capacity if not given a periodic full discharge cycle.
- **Nickel Metal Hybride (Ni-MH):** Nickel metal hybride has become one of the most readily available rechargeable batteries for consumer use and is used for the most at the same applications as Ni-Cd. It provides 40 percent higher specific energy than the standard Ni-Cd. Good ventilation is important in the room to avoid explosive concentration of hydrogen. The battery is more delicate and trickier to charge than Ni-Cd. High self-discharge is of ongoing concern, and devices with a Ni-MH battery gets “flat” when put away for only a few weeks.
- **Sodium Sulphur (Na-S):** The sodium sulphur batteries, are manufactured from cheap and plentiful raw materials and has higher specific energy compared to Li-ion batteries. However, the manufacturing processes and the need for insulation, heating and thermal management make these batteries quite expensive and counteracts the benefits, because these type of batteries need molten sodium-metal as a anode, making it necessary to operate at 300°C.
- **Lithium-ion (Li-ion):** Lithium-ion batteries can consist of different material and chemistries in the electrodes and the electrolyte, as well as manufacturing processes and related materials.

Has the highest specific energy of commercially available batteries and highest energy density of commercially available batteries. Some disadvantages are flammable electrolyte and potentially limited availability of materials. Among the different **cathodes** and **anodes** of lithium-ion batteries are shown in the table 1.2:

Table 1.2: Lithium-ion batteries types [19].

Lithium-ion Battery	
Cathode Chemistries	Anode Chemistries
Nickel Manganese Cobalt Oxide (NMC)	Graphene
Nickel Cobalt Aluminium (NCA)	Titanate
Lithium Cobalt Oxide (LCO)	Silicon
Lithium Manganese Oxide Spinel (LMO)	
Lithium Iron Phosphate (LFP)	

- **Li-polymer:** Also known like *solid-state battery*, this type of technology gives freedom in design of the battery geometry and improvement of the packing efficiency of the cells. It facilitates a long cycle life and offers the possibility of employing high-voltage cathodes. All these effects increase the practical battery energy density. The biggest challenge however is regarded as the volume changes of the electrodes during charging and discharging, that causes loss of contact between the electrode and the electrolyte. These effects all make ion conductivity low. A polymer interlayer between the electrodes and the solid electrolyte has been proposed to overcome these challenges. Some disadvantages of this technology are its high production cost and low lifetime.
- **Lithium-Sulphur (Li-S):** Like conventional lithium-ion batteries, these batteries use Li⁺ ions as energy carriers. They react with sulfur at the cathode giving some high theoretical values for specific energy and energy density. Hence these batteries will be very attractive for marine applications. Safety concerns related to these batteries, like dendrite formation, have been improved. However, this compromises the specific energy and energy density and increases the cost dramatically. Other obstacles to overcome are high electrical resistance, capacity fading, self-discharge, mainly due to the so-called shuttle effect.

About the different types of batteries described above, there is a technical comparison in the table 1.3.

Table 1.3: Energy Storage Systems Comparison [21]

Battery	Energy Density (Wh/kg)	Volumetric Energy Density (Wh/L)	Cost (USD/kWh)	Lifecycle	Efficiency (%)
Lead-Acid	20-40	60-110	165	500-1000	50-95
Ni-Cd	20-60	60-150	-	2000-2500	70-90
Ni-MH	50-70	140-300	146	500-2000	89-92
Na-S	~120	150-300	400	>1500	86-92
Li-ion	50-250	250-675	200-600	400-9000	85-99
Li-polymer	~150	250-730	-	-	-
Li-S	400	350	400-500	1500	-

Different ships have different operating profiles and batteries must respond to specific energy and power demand while also having in consideration the desired/expected life-cycle for the battery [19]. In this scenario, lithium batteries are superior to the other mentioned batteries in terms of energy density, lifecycle, and efficiency [21]. Lithium batteries have been gaining traction throughout the years, with many applications including portable electronics such as mobile phones, and automobile applications such as electric vehicles. Also are slowly being adopted for aerospace, military and marine applications. Lithium batteries are highly versatile energy storage devices that can be altered to meet specific demands, making them one of the best choices for marine application.

Considering the incorporation of batteries into the propulsion system of vessels, different propulsion architectures are recognized that provide a different use for batteries, even without considering them, as is the case with conventional propulsion architecture [19] [20]. This can be seen in the figures 1.9, 1.10 and 1.11.

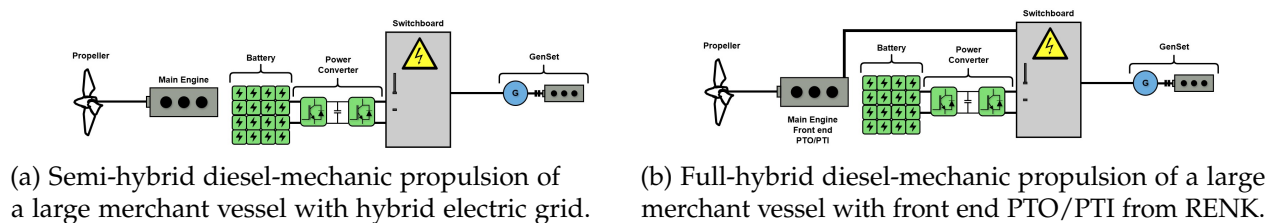
a) Diesel-mechanic engine conventional system



Figure 1.9: Traditional diesel-mechanic propulsion of a large merchant vessel.

In general, the configuration shown in Figure 1.9 is the most common installation on large ocean-going vessels [20]. It consists of a two-stroke main engine responsible for propulsion, while the ship's load is covered by auxiliary engines/sensets installed on board the vessel.

b) Diesel-mechanic engine hybrid system



(a) Semi-hybrid diesel-mechanic propulsion of a large merchant vessel with hybrid electric grid.

(b) Full-hybrid diesel-mechanic propulsion of a large merchant vessel with front end PTO/PTI from RENK.

Figure 1.10: Semi-hybridization.

In Fig. 1.10a, a battery is included in a traditional system where the electric grid is separated from the propulsion of the vessel [20]. The battery will in such a solution be effective for smoothing the connected hotel electrical loads and contribute to handle large load steps or peaks [19]. When large load steps are increased, the number of auxiliary motors also increases. In some cases, the load can regenerate energy, for example, in crane operations. In such cases, the battery can be

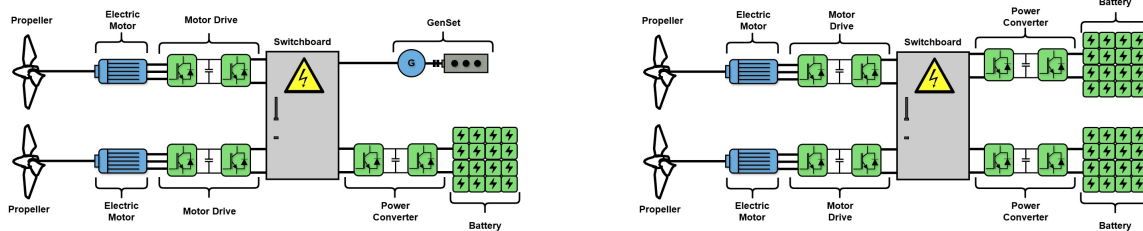
used to capture the energy.

In 1.10b, a battery is included in combination with a shaft generator power take-out (PTO) and power take-in (PTI) [20]:

- **Power Take-Out (PTO):** The electrical/mechanical hybrid solution allows electricity to be generated by the main engine.
- **Power Take-In (PTI):** The electrical/mechanical hybrid solution allows propulsion power to be produced by generator sets and batteries. Additional power is possible (boost mode) when the main engine and PTI motor are running in parallel.

In addition, a ship with a hybrid electric/mechanical propulsion system may have a direct current distribution system, which can adjust the speed of the primary generator engines to the optimum fuel level depending on the load. This will reduce fuel consumption and minimize environmental impact [19].

c) Hybrid/electric propulsion system with electric motors



(a) Full-hybrid diesel-electric propulsion with batteries in a hybrid system.

(b) Pure electric propulsion with batteries as the primary energy source.

Figure 1.11: Propulsion system based on electric motors, with different power sources.

A full-hybrid diesel-electric solution, as is shown in fig. 1.11a, reduces the noise and vibration level on the ship [19]. Supplementary to be a source of energy for propulsion, the batteries will also smooth the load variations on the generator sets. This type of solution can for instance facilitate the use of zero emission operation when entering a harbour. Another version of this solution is with the batteries distributed into the drives directly [19]. In this way, each propulsion unit is independent of a common power source, which could be valuable for vessels that require highly reliable propulsion thrust.

In the fig. 1.11b, the propellers are connected to electric motors, which are driven by the energy stored in a battery system that is typically charged from shore [20]. In some battery-electric systems, a smaller diesel generator is sometimes included to ensure operation if the batteries fail to charge or to enable longer voyages, i.e. when the vessel has to dock [20]. For such a solution the batteries will be charged through an AC/DC converter. The converter can either be located on board the vessel or on shore. The fig. 1.11b shows two independent battery systems that deliver power to the thruster. This is according to class rules, which require that two independent battery systems are installed to provide propulsion power in case one of the systems fail [19].

Electrification of ship's propulsion is increasingly recognised as a core part of the maritime industry's future [23]. According to [26], the adoption of alternative fuels in the global fleet by number of ships is dominated by Liquefied Natural Gas (LNG) together with hybrid/battery ships, as is shown in Table 1.4.

Table 1.4: Ships 2022.

Alternative Fuel Ships		
	In operation during 2022	Ordered
LNG	926	534
Hybrid/battery powered ships	396	417
Total	1349	1046

In Table 1.4, can be seen that the proportion of hybrid/battery propulsion systems has been increasing since 2022. However, it should be noted that the use of batteries as a source of propulsion for ships is limited to certain types of ships, the main ones being ferries, dynamic positioning ships and platforms, dredging, short-range ships, wind farm support vessels, and tugs. For ships that travel most of the time at constant speed, such as container ships, it make no sense to replace a diesel mechanical drive by hybrid concepts, because the diesel-mechanical drives are optimized for this operation and then getting a lower consumption than a diesel electric or hybrid system [4].

Harbour tugs are a sector of the maritime industry with great potential for the implementation of hybrid propulsion, as they experience significant variations in power demand during operations [5].

Ports are imposing more and more stringent regulations on exhaust and noise emissions. Those demands for cleaner operation can be easily met by the implementation of battery energy storage systems (BESS) (with hybrid-electric propulsion). Using a combination of electric power, BESS and combustion engines, a hybrid tug optimizes engine loading, resulting in lower specific fuel consumption, higher efficiency, lower emissions and lower fuel consumption.

1.1.5 Tugboats



Figure 1.12: Tugboat

Tugboats, as can be seen in fig. 1.12, are ideal candidates for hybridization due to their operational profiles, which involve prolonged idling, low-speed maneuvering, and frequent speed changes that lead to inefficient fuel consumption [21]. Among vessel types, tugboats—used for towing and maneuvering large ships—are among the highest emitters per unit of energy due to their highly variable load profiles [4], since they spend a lot of time at 10-20% load (inefficient for diesel engines) and very short peaks at 100%. This technically justifies the use of batteries.

1.2 Challenges and Research Opportunities

Several studies have examined hybrid propulsion systems [8], identifying the main topics of study related to hybridization, which are listed below [23]:

- Optimal sizing of the batteries.
- Optimal control strategy for the parallel operation of a battery and diesel generators.
- Control and safe operation of lithium battery packs.
- BESS control system architecture.
- Design of a battery management system (BMS).
- Safety assessment of the BESS, its elements and safeguarding measures required by the classification societies.
- Safety assessment of the BESS, its elements and safeguarding measures required by the classification societies.

- Degradation of Li-ion batteries.
- Battery application for large vessels.
- Reduction of fuel consumption and emissions.

Some studies related to hybridization are listed below, with emphasis on those related to tugboat hybridization:

- The feasibility of hybrid propulsion concepts is examined for different types of vessels, including tugboats [4], showing that there are definitely opportunities to save energy by using hybrid drive systems on ships, even return the much higher investment by savings in fuel.
- A genetic algorithm to obtain the optimal operating points regard with series hybrid configurations in power generation, especially for the case of heavy duty hybrid drive applications [18].
- A marine hybrid propulsion system, focusing on vector control of the electric motor during different modes and verifying the control feasibility [22].
- An analytical design methodology for BESS in hybrid marine vessels [24].
- The optimization of hybrid propulsion system design for a tugboat has been explored, presenting a methodology that streamlines powertrain component sizing and control, minimizing costs for a specific operating profile [27].
- A coordinated control strategy for a variable-speed hybrid tugboat have been presented to improve fuel economy. The proposed strategy, validated through simulations and a smallscale experimental testbed, showed reduced costs and lower CO2 emissions [28].

However, the challenge lies in finding a clear and detailed methodology for the hybridization of vessels such as tugboats that provides a path from the conceptual phase to the technical details and subsequent implementation, taking into account technical and economic aspects.

1.3 Thesis Objectives and Outline

The general objective of this work is to develop a methodology for the design and sizing of the battery bank for a hybrid tugboat, which will allow us to determine the benefits that would be obtained from hybridization, analyzing the challenge of high CAPEX (initial cost) of batteries versus savings in OPEX (fuel/maintenance).

Specific objectives include:

- Model the operation of a conventional tugboat
- Define hybrid architecture.
- Propose the service power of the electrical system.
- Preliminary design of the battery bank based on technical and economic criteria.
- Simulation of the proposed cell and battery bank.

1.4 Methodology

The methodology to follow for the development of this work is:

- **State of Art:** Review of existing hybrid propulsion system types and battery technologies.
- **Mathematical Analysis and Modeling:** Theoretical study of the tugboat's load profile and the implementation of batteries to support the vessel's operation.
- **Results:** Simulation of the models obtained through Plecs or MatLab to complete the conceptual analysis and analyze the results related to the battery bank.

1.5 Scope and Limitations

The limitations of the work are:

- **Architecture Layout:** Only one hybridization layout will be analyzed and compared with the tugboat's conventional propulsion system.
- **Bill of Components (BOM):** Within the list of components of the hybrid propulsion system, the associated cost of each will be determined, estimating the costs of those components for which no commercial information is available.
- **Economic analysis:** Although information on costs related to the hybrid propulsion system will be added, it should be noted that inflation or deflation will not be taken into account in this exercise. Furthermore, this economic assessment does not consider the cost of the port charger.
- **Simulation considerations:** Based on the data provided on the tugboat's operational load profile, simulations will be performed on the sized battery bank. For power converter simulations, simulations will be performed with a shorter duration, given the amount of data required to analyze the battery bank's discharge profile. The thermal effects of the semiconductors used shall be disregarded.

In conclusion, this work will contribute to the analysis of the different initial stages of hybridizing a tugboat, considering the essential technical aspects that allow us to understand the benefits of this conversion, with an emphasis on the electrical stage, specifically the batteries.

1.6 Chapter Summary

Chapter 2

Identification of essential technical and operational characteristics of a tugboat and analysis of the associated operational load profile.

Chapter 3

Hybridization analysis and new operational services that a hybrid tugboat can offer based on its cargo operational profile.

Chapter 4

Design and sizing of battery banks, considering different technologies, technical variables, and economic analysis.

Chapter 5

Simulation of the battery bank selected for the vessel, considering the mathematical model of a cell.

Chapter 2

Design of the battery bank for a hybrid propulsion system

This chapter presents the preliminary design process for the battery bank required for the tugboat's hybrid propulsion system. The energy capacity, power requirements, and operating constraints are determined, and then different storage technologies are evaluated from a technical and economic perspective.

2.1 Energy Capacity (kWh)

To calculate the net capacity required for the battery bank design, various factors that provide context must be considered [65] [66].

The fundamental energy requirement, denoted as E_{load} is calculated by integrating the power demand over the duration of the design mission.

$$E_{\text{load}} = \int_0^t (P_{\text{propulsion}}(t) + P_{\text{hotel}}(t)) dt$$

For discrete phases, this summation becomes:

$$E_{\text{load}} = \sum_{i=1}^n P_i \times t_i$$

Where:

- P_i is the total electrical load at the DC bus (propulsion + hotel) in kilowatts (kW) for phase i .
- t_i is the duration of phase i in hours.

This value, E_{load} , represents the energy that must be delivered to the propellers and the hotel loads. However, calculating the installed battery capacity ($E_{\text{installed}}$) requires expanding this value to account for the physical and chemical realities of the energy storage system.

This transformation is achieved through the application of specific sizing factors.

a) System Efficiency Factor (η_{sys})

Energy stored in the electrochemical cells must traverse a electrical path to reach the propulsion motor. Each component in this chain extracts an efficiency penalty (heat loss).

1. **Coulombic/Voltaic Efficiency (η_{bat}):** Li-ion batteries are highly efficient, but internal resistance (R_{int}) causes I^2R losses during discharge. Typical round-trip efficiency is 92 – 96%; discharge efficiency is generally taken as 0.96 to 0.98 [69].
2. **DC/DC Converters η_{conv} :** Many hybrid architectures utilize DC/DC converters to decouple the battery voltage from the main DC bus voltage. These typically operate at 0.97 to 0.98 efficiency.
3. **Variable Frequency Drives (VFD) η_{vfd} :** This inverters for electric motors converting DC bus power to AC for the motors operate at 0.96 to 0.98 efficiency.
4. **Electric Motor η_{motor} :** Permanent Magnet (PM) machines are preferred for hybrid tugs due to their high efficiency at partial loads compared to induction motors. PM motor efficiency is typically 0.95 to 0.97.

To size the battery based on shaft power demand, the cumulative efficiency is:

$$\eta_{sys} = \eta_{bat} \times \eta_{conv} \times \eta_{vfd} \times \eta_{motor} \quad (2.1)$$

b) Maximum Depth of Discharge (DOD_{max}) and Reserve Margins

The Depth of Discharge (DOD) is the percentage of the battery's capacity that has been discharged relative to its total capacity. While Li-ion chemistries can be discharged to 100% (0% SOC), doing so drastically reduces cycle life and increases the risk of cell degradation.

- **Cycle Life Correlation:** The relationship between DOD and cycle life is non-linear. For example, an NMC cell might offer 2,000 cycles at 100% DOD but 5,000 cycles at 80% DOD.
- **Operational Reserve:** Marine safety standards often dictate a "reserve energy" capability. For a tugboat, this might mean retaining enough energy after a standard operation to perform emergency maneuvering or firefighting for a specified duration.
- **Voltage Sag:** At very low SOC (< 10%), the cell voltage drops rapidly. To maintain the required bus voltage and power delivery, the current must increase, leading to higher thermal stress.

Standard marine practice utilizes a Maximum DOD of 80%. This implies the battery operates between 10% and 90% SOC.

$$\text{Factor}_{DOD} = \frac{1}{DOD_{max}} \quad (2.2)$$

c) Battery Aging Factor (k_{aging})

Batteries degrade over time due to two primary mechanisms:

1. **Calendar Aging:** Loss of capacity due to chemical decomposition of the electrolyte and SEI (Solid Electrolyte Interphase) layer growth, occurring even when the battery is idle. This is temperature and SOC dependent.
2. **Cyclic Aging:** Mechanical stress on the electrode materials during ion insertion/extraction, leading to cracking and active material loss. This is driven by throughput (Ah) and C-rates.

End of Life (EOL) Definition: In the maritime industry, a battery system is considered to have reached its End of Life (EOL) when its capacity fades to 80% of its original nominal capacity (Beginning of Life, BOL). The vessel must be capable of performing its design mission on the last day of the battery's life (e.g., Year 10). Therefore, the battery must be oversized at installation so that the faded capacity still meets the requirement.

$$k_{aging} = \frac{1}{SOH_{EOL}} \quad (2.3)$$

d) Temperature Correction Factor (k_{temp})

Battery performance is inextricably linked to temperature. The standard rating is typically given at 25°C.

- **Low Temperature Effects:** At temperatures below 15°C, the viscosity of the electrolyte increases, and ion mobility decreases. This manifests as increased internal resistance. At 0°C, a Li-ion battery may only be able to deliver 80% to 90% of its rated capacity, and its ability to accept charge (regenerative braking) is severely curtailed to prevent lithium plating.
- **High Temperature Effects:** Operating above 35-40°C accelerates the chemical side reactions that drive aging. While capacity might initially increase slightly due to lower resistance, the long-term penalty is severe lifespan reduction.

Tugboat battery rooms are invariably climate-controlled (Liquid Cooling + HVAC) to maintain an optimal window of 20°C–25°C.

A conservative Temperature Correction Factor of 1.05 to 1.10 (5 – 10% oversizing) is applied. This ensures that the rated energy is available even if the ambient conditions in the battery room deviate from the ISO standard day.

e) Capacity Rating Factor and Peukert Efficiency (k_{c-rate})

Peukert's Law ($C_p = I^k t$) describes how the available capacity of a battery decreases as the discharge rate increases. For Lithium-ion batteries, the effect is minimal, with k values typically between 1.01 and 1.05.

- **Low Power Applications:** For hotel loads (< 0.5C), the factor is negligible (~ 1.0).
- **High Power Applications:** For propulsion boost (> 2C), a derating factor is prudent.

To calculate this factor, an initial C_{rate} can be made based on the required energy E_{load} and the service power $P_{service}$ using the following formula [20]:

$$C_{rate}[h^{-1}] = \frac{E_{load}[kWh]}{P_{service}[kWh]} \quad (2.4)$$

This provides an approximation of the C_{rate} to determine the value of the k_{c-rate} factor.

2.1.1 Battery Sizing

Based on the factors described previously, the calculations used to obtain the total installed energy $E_{installed}$ are presented:

1. **Total Load:** This corresponds to adding the energy from the load profile E_{prof} to the energy used for hotel loads E_{hotel}

$$E_{load} = E_{prof} + E_{hotel}$$

2. **C-rate:** Initial calculation of the C_{rate} to determine the associated factor k_{c-rate}

$$C_{rate} = \frac{E_{load}}{P_{service}} < 0 \Rightarrow k_{c-rate} = 1$$

$$C_{rate} = \frac{E_{load}}{P_{service}} \geq 0 \Rightarrow k_{c-rate} > 1$$

3. **System Efficiency:** To supply power to the mechanical shafts, the battery must generate

$$E_{batt.out} = E_{load} / \eta_{sys}$$

4. **Max DOD:** Ensure that the battery is only used to the appropriate depth of discharge relative to its initial nominal capacity (beginning of life, BOL). To do this, calculate

$$E_{BOL} = E_{batt.out} / DOD_{max}$$

5. **Aging Factor:** To ensure that the battery bank has a long service life, the aging factor must be taken into account

$$E_{EOL} = E_{BOL} / k_{aging}$$

6. **Apply Temperature/Rating Factors:** Consider the variation in temperature and battery discharge rate through

$$E_{installed} = E_{EOL} \cdot k_{temp} \cdot k_{c-rate}$$

The following equation is presented to obtain the total installed capacity based on the factors explained above.

$$E_{installed} = \frac{E_{load} \cdot k_{temp} \cdot k_{c-rate}}{DOD_{max} \cdot k_{aging} \cdot \eta_{sys}} \quad (2.5)$$

Based on equation (2.5), the factors used for sizing both battery banks proposed for hybridization are presented.

Table 2.1: Energy Factors

Energy Factors	
η_{sys}	0.88
DOD_{max}	0.8
k_{aging}	0.8
k_{temp}	1.05

Based on the information shown in Figures ??, ??, ?? and ??, it can be seen that one maneuver demands the most power from the battery bank in both scenarios. Thus, the total installed power for the two different battery bank options can be calculated. This can be seen in Table 2.2.

Table 2.2: Comparison of Propulsion Systems in Tugboats and Performance.

	BESS 15% P_{nom}	BESS 25% P_{nom}
E_{prof}	1,278[kWh]	3,102[kWh]
E_{hotel}	192[kWh]	192[kWh]
E_{load}	$\approx 1,470$[kWh]	$\approx 3,300$[kWh]
$P_{service}$	600[kW]	1,000[kW]
C_{rate}	≈ 0.4[h⁻¹]	≈ 0.3[h⁻¹]
k_{c-rate}	1[-]	1[-]
$E_{bat.out}$	1,670[kWh]	3,750[kWh]
E_{BOL}	2,088[kWh]	4,688[kWh]
E_{EOL}	2,610[kWh]	5,859[kWh]
$E_{installed}$	$\approx 2,800$[kWh]	$\approx 6,300$[kWh]

2.2 Nominal Battery Voltage

The system voltage should be determined by the cost, desired generator and propulsion motor drive voltage, converter design, load requirements, cable and bus-bar rating, and the fault energy [29]. Generally, low-voltage dc (LVdc) systems are lower than 1 kV, with power system capacity in the range of 1–20MW, while medium-voltage dc (MVdc) systems are higher than 1 kV, with 20–100 MW.

- **750V DC:** This range is common in earlier hybrid designs, smaller tugboats, or retrofits where space for series-connected modules might be limited [72]. It balances safety requirements with the need for power density but becoming less favorable for tugs with MW-scale power requirements due to the heavy cabling required.
- **1000V DC:** This is the emerging standard for high-power vessels. Higher voltage reduces the current required for the same power ($P = VI$) and copper losses ($P_{loss} = I^2R$).

The battery bank is constructed of modules in series to achieve the voltage.

$$N_{series} = V_{bus.req} / V_{cell.nom} \quad (2.6)$$

For the battery bank in both options, a nominal voltage of $V_{bus.req} = 750$ [V] will be proposed.

2.3 Battery technologies applicable to tugboats

Although several battery technologies for different vessels were mentioned in section 1.1.4, in the field of hybrid and electric tugboats, three main lithium-ion chemical compositions dominate this sector [68]:

- **Nickel Manganese Cobalt (NMC)**,
- **Lithium Iron Phosphate (LFP)**,
- **Lithium Titanate Oxide (LTO)**.

2.3.1 Nickel Manganese Cobalt (NMC)

NMC technology (LiNiMnCoO_2) is preferred for electric vehicles and within the maritime industry as its life cycle is long while the energy density is satisfying [20].

NMC has historically been the chemistry of choice for first-generation hybrid vessels due to its high specific energy.

Main features on NMC include the following:

- **Energy Density:** NMC cells offer between 150 and 220 Wh/kg, which translates into lower densities after taking into account the BMS, cooling, and rack [67].
- **C-rate:** Standard marine NMC modules (e.g., Corvus Orca Energy) support high discharge rates, typically 3C continuous [71], ideal for the high power density demands of a tugboat.
- **Cycle Life:** NMC typically offers between 1000 and 1500 cycles [76], which may be sufficient for operating modes where the battery is used for peak shaving rather than performing full cycles every day.
- **Cost:** Cell prices range from US \$120 – US \$150/kWh [67]. The main drawback is thermal stability; NMC has a lower thermal runaway temperature, requiring robust active cooling systems that could increase its cost [81].

2.3.2 Lithium Iron Phosphate (LFP)

LFP (LiFePO_4) is rapidly gaining market share in the maritime sector, driven by safety and longevity.

- **Energy Density:** The cell density is 90-160 Wh/kg [76]. Tugboats require considerable ballast to achieve the draft necessary for propeller immersion. LFP batteries could replace lead or iron ballast, serving a dual function as a “**stability weight**” [67], representing around 30% of the total ballast, which could reach approximately 55[m³] [70].
- **Safety:** The simple chemistry of LFP batteries (no nickel or cobalt) makes them safe and robust. These batteries are used in applications where safety and long life are more important than compact size [82].
- **Cycle Life:** The LFP offers between 2,000 and more than 5,000 cycles, even with a high discharge depth [76]. Some sources suggest that it lasts between 2 and 3 times longer than the NMC [67], making it ideal for applications that require longer life and reliability.
- **Cost:** The LFP has an estimated cell cost of between US \$80 – US \$100/kWh, making it 30% cheaper than the NMC option in 2026 [67].

2.3.3 Lithium Titanate Oxide (LTO)

Lithium titanate batteries ($\text{Li}_4\text{Ti}_5\text{O}_{12}$) are the only ones with a unique anode surface made from lithium and titanium oxides. These batteries for electric boats, also known as LTO batteries, operate in extreme weather conditions, offering greater safety [81].

- **Performance:** One of the main advantages of LTO is its high C-rate capacity (i.e., high maximum power) [89]. It offers exceptional service life (more than 10,000 cycles) [76].
- **Drawbacks:** LTO has the lowest energy density (60–120 Wh/kg) and the highest cost [76].
- **Cycle Life:** LTO batteries can operate safely for 25,000 cycles, achieving a service life of more than 20 years [89].

2.4 Battery Bank Design

Taking into account the battery technologies used in marine environments, the total energy required by the operating profile, and the voltage of the battery bank, the steps to follow to obtain the total number of cells needed to find the right battery bank are shown below [73].

To this end, the selected cells shall be considered to have the following input parameters, as provided by the manufacturer. This is shown in Table 2.3.

Table 2.3: Technical input information.

Input technical information	
V_{bc}	Battery cell voltage [V]
E_{bc}	Battery cell energy [Wh]
C_{bc}	Battery cell capacity [Ah]
$C\text{-rate}_{bcc}$	Continuous C-rate of the battery cell (discharge) [h^{-1}]
$C\text{-rate}_{bpc}$	Peak C-rate of the battery cell [h^{-1}]
m_{bc}	Battery cell mass [kg]
vol_{bc}	Battery cell volume [m^3]

In addition, the calculated energy and the proposed nominal voltage for each bank to be designed shall be considered as inputs. Below is the information that will be used for the small battery bank and the large battery bank.

- Small Battery Bank (15% P_{nom}):
 - V_{bp} (Battery pack voltage): 750 [V].
 - E_{bp} (Battery pack total energy): 2,800[kWh].
- Large Battery Bank 25% P_{nom}):
 - V_{bp} (Battery pack voltage): 750 [V].
 - E_{bp} (Battery pack total energy): 6,300[kWh].

The battery bank will be considered to consist of several strings connected in parallel, each of which consists of several cells connected in series. Also, the volume of the battery pack will be calculated taking into account only the battery cells, without considering other factors such as electronic circuits, cooling circuit, battery casing, wiring, etc.

Next, we will present the basic calculations used to characterize the battery bank. These include:

- Essential electrical parameters.
- Electrical performance parameters.
- Mechanical parameters.

2.4.1 Calculation of essential electrical parameters

The number of battery cells connected in series $N_{sc}[-]$ in a string is calculated by dividing the nominal battery pack voltage $V_{bp}[V]$ to the voltage of each battery cell $V_{bc}[V]$. The number of strings must be an integer. Therefore, the result of the calculation is rounded to the higher integer.

$$N_{sc} = \frac{V_{bp}}{V_{bc}} \quad (2.7)$$

The energy content of a string $E_{bs}[Wh]$ is equal with the product between the number of battery cells connected in series $N_{sc}[-]$ and the energy of a battery cell $E_{bc}[Wh]$.

$$E_{bs} = N_{sc} \cdot E_{bc} \quad (2.8)$$

Total number of strings of the battery pack $N_{sb}[-]$ is calculated by dividing the battery pack total energy $E_{bp}[Wh]$ to the energy content of a string $E_{bs}[Wh]$. The number of strings must be an integer. Therefore, the result of the calculation is rounded to the higher integer.

$$N_{sb} = \frac{E_{bp}}{E_{bs}} \quad (2.9)$$

The battery pack capacity $C_{bp}[Ah]$ is calculated as the product between the number of strings $N_{sb}[-]$ and the capacity of the battery $C_{bc}[Ah]$.

$$C_{bp} = N_{sb} \cdot C_{bc} \quad (2.10)$$

Total number of cells of the battery pack $N_{bp}[-]$ is calculated as the product between the number of strings $N_{sb}[-]$ and the number of cells in a string $N_{sc}[-]$.

$$N_{bp} = N_{sb} \cdot N_{sc} \quad (2.11)$$

2.4.2 Calculation of electrical performance parameters

The string continuous current $I_{scc}[A]$ is the product between the continuous C-rate of the battery cell C-rate_{bcc} $[h^{-1}]$ and the battery cell capacity $C_{bc}[Ah]$.

$$I_{scc} = \text{C-rate}_{bcc} \cdot C_{bc} \quad (2.12)$$

The battery pack continuous current $I_{bpc}[A]$ is the product between the string continuous current $I_{scc}[A]$ and the number of strings of the battery pack $N_{sb}[-]$.

$$I_{bpc} = I_{scc} \cdot N_{sb} \quad (2.13)$$

The battery pack continuous power $P_{bpc}[W]$ is the product between battery pack continuous current $I_{bpc}[A]$ and the battery pack voltage $V_{bp}[V]$.

$$P_{bpc} = I_{bpc} \cdot V_{bp} \quad (2.14)$$

The string peak current $I_{spc}[A]$ is the product between the peak C-rate of the battery cell C-rate $_{bpc}[h^{-1}]$ and the battery cell capacity $C_{bc}[Ah]$.

$$I_{spc} = C\text{-rate}_{bpc} \cdot C_{bc} \quad (2.15)$$

The battery pack peak current $I_{bpp}[A]$ is the product between the string peak current $I_{spc}[A]$ and the number of strings of the battery pack $N_{sb}[-]$.

$$I_{bpp} = I_{spc} \cdot N_{sb} \quad (2.16)$$

The battery pack peak power $P_{bpp}[W]$ is the product between battery pack peak current $I_{bpp}[A]$ and the battery pack voltage $V_{bp}[V]$.

$$P_{bpp} = I_{bpp} \cdot V_{bp} \quad (2.17)$$

2.4.3 Calculation of mechanical parameters

The battery pack mass (cells only) $m_{bp}[kg]$ is the product between the total number of cells $N_{bp}[-]$ and the mass of each battery cell $m_{bc}[kg]$.

$$m_{bp} = N_{bp} \cdot m_{bc} \quad (2.18)$$

The volume of the battery pack (cells only) $vol_{bp}[m^3]$ is the product between the total number of cells $N_{bp}[-]$ and the volume of each battery cell $vol_{bc}[m^3]$. This volume is only used to estimate the final volume of the battery pack, since it does not take into account the auxiliary components/systems of the battery [73].

$$vol_{bp} = N_{bp} \cdot vol_{bc} \quad (2.19)$$

2.4.4 Calculated NMC battery banks

Three different models of NMC batteries and their technical specifications are presented in Table 2.4.

Table 2.4: Technical specification of NMC Batteries.

NMC Batteries			
	ABL-012010P	ABL-012030P	E4050T1
V_{bc}	12[V]	12[V]	2.84[V]
E_{bc}	120[Wh]	360[Wh]	264[Wh]
C_{bc}	10[Ah]	30[Ah]	128[Ah]
$C\text{-rate}_{bcc}$	1 [h ⁻¹]	1 [h ⁻¹]	1 [h ⁻¹]
$C\text{-rate}_{bpc}$	3 [h ⁻¹]	3 [h ⁻¹]	3 [h ⁻¹]
m_{bc}	0.8[kg]	2.1[kg]	0.48[kg]
vol_{bc}	$0.49 \cdot 10^{-3}[\text{m}^3]$	$1.13 \cdot 10^{-3}[\text{m}^3]$	$4.3 \cdot 10^{-3}[\text{m}^3]$

The technical specifications of each model, in addition to the input parameters of the battery bank, allow the necessary calculations to be made to characterize both small and large NMC battery banks.

a) NMC Small Battery Bank (Battery Bank 15% P_{nom})

The essential electrical characteristics, performance parameters, and physical characteristics of the small NMC battery bank are presented below in Tables 2.5 and 2.6, respectively.

Table 2.5: Essential parameters of the small NMC battery bank.

Essential parameters, NMC battery bank 15% P_{nom}			
	ABL-012010P	ABL-012030P	E4050T1
N_{sc}	63	63	264
E_{bs}	7.56[kWh]	22.68[kWh]	113[kWh]
N_{sb}	371	124	25
C_{bp}	3,710[Ah]	3,720[Ah]	3,200[Ah]
N_{bp}	23,373	7,812	6,600

Table 2.6: Performance parameters of the small NMC battery bank.

Performance parameters, NMC battery bank 15% P_{nom}			
	ABL-012010P	ABL-012030P	E4050T1
I_{scc}	5[A]	30[A]	128[A]
I_{bpc}	1.86[kA]	3.72[kA]	3.2[kA]
P_{bpc}	1,391[kW]	2,790[kW]	2,400[kW]
I_{spc}	20[A]	60[A]	750[A]
I_{bpp}	7.42[kA]	7.44[kA]	18.75[kA]
P_{bpp}	5,565[kW]	5,580[kW]	14,062[kW]
m_{bp}	18,698[kg]	16,405[kg]	3,143[kg]
vol_{bp}	11.45[m ³]	8.80[m ³]	28.11[m ³]

b) NMC Large Battery Bank (Battery Bank 25% P_{nom})

The essential electrical characteristics, performance parameters, and physical characteristics of the large NMC battery bank are presented below in Tables 2.7 and 2.8, respectively.

Table 2.7: Essential parameters of the large NMC battery bank.

Essential parameters, NMC battery bank 25% P_{nom}			
	ABL-012010P	ABL-012030P	E4050T1
N_{sc}	63	63	264
E_{bs}	7.56[kWh]	22.68[kWh]	113[kWh]
N_{sb}	834	278	56
C_{bp}	8,340[Ah]	8,340[Ah]	7,168[Ah]
N_{bp}	52,542	17,514	14,784

Table 2.8: Performance parameters of the large NMC battery bank.

Performance parameters, NMC battery bank 25% P_{nom}			
	ABL-012010P	ABL-012030P	E4050T1
I_{sc}	5[A]	30[A]	128[A]
I_{bpc}	4.17[kA]	8.34[kA]	7.17[kA]
P_{bpc}	3,128[kW]	6,255[kW]	5,376[kW]
I_{spc}	20[A]	60[A]	750[A]
I_{bpp}	16.68[kA]	16.68[A]	42[kA]
P_{bpp}	12,510[kW]	12,510[kW]	31,500[kW]
m_{bp}	40,034[kg]	36,779[kg]	7,096[kg]
vol_{bp}	25.75[m ³]	19.73[m ³]	28.11[m ³]

2.4.5 Calculated LFP battery banks

As in the previous case, Table 2.9 shows three different battery models, but now LFP, with their technical specifications.

Table 2.9: Technical specification of LFP Batteries.

LFP Batteries			
	URB1270	PSL-SC-12500	U27-36XP
V_{bc}	12.8[V]	12.8[V]	38.4[V]
E_{bc}	97[Wh]	640[Wh]	1,920[Wh]
C_{bc}	7.6[Ah]	50[Ah]	50[Ah]
$C\text{-rate}_{bcc}$	2 [h ⁻¹]	1 [h ⁻¹]	1 [h ⁻¹]
$C\text{-rate}_{bpc}$	7 [h ⁻¹]	3 [h ⁻¹]	3 [h ⁻¹]
m_{bc}	1[kg]	6.3[kg]	18.7[kg]
vol_{bc}	$0.9 \cdot 10^{-3}$ [m ³]	$5.59 \cdot 10^{-3}$ [m ³]	$11.84 \cdot 10^{-3}$ [m ³]

These technical specifications for each model enable the necessary calculations to be made to characterize both small and large LFP battery banks.

a) LFP Small Battery Bank (Battery Bank 15% P_{nom})

The essential electrical characteristics, performance parameters, and physical characteristics of the small LFP battery bank are presented below in Tables 2.10 and 2.11, respectively.

Table 2.10: Essential parameters of the small LFP battery bank.

Essential parameters, LFP battery bank 15% P_{nom}			
	URB1270	PSL-SC-12500	U27-36XP
N_{sc}	59	59	20
E_{bs}	5.72[kWh]	37.76[kWh]	38.40[kWh]
N_{sb}	490	75	73
C_{bp}	3,724[Ah]	3,750[Ah]	3,650[Ah]
N_{bp}	28,910	4,425	1,460

Table 2.11: Performance parameters of the small LFP battery bank.

Performance parameters, LFP battery bank 15% P_{nom}			
	URB1270	PSL-SC-12500	U27-36XP
I_{sc}	15[A]	50[A]	50[A]
I_{bpc}	7.35[kA]	3.75[kA]	3.65[kA]
P_{bpc}	5,513[kW]	2,813[kW]	2,738[kW]
I_{spc}	55[A]	140[A]	150[A]
I_{bpc}	26.95[kA]	10.5[kA]	10.95[kA]
P_{bpc}	20,212[kW]	7,875[kW]	8,213[kW]
m_{bp}	28,910[kg]	27,877[kg]	27,302[kg]
vol_{bp}	26.27[m ³]	24.73[m ³]	17.28[m ³]

b) LFP Large Battery Bank (Battery Bank 25% P_{nom})

The essential electrical characteristics, performance parameters, and physical characteristics of the large LFP battery bank are presented below in Tables 2.12 and 2.13, respectively.

Table 2.12: Essential parameters of the large LFP battery bank.

Essential parameters, LFP battery bank 25% P_{nom}			
	URB1270	PSL-SC-12500	U27-36XP
N_{sc}	59	59	20
E_{bs}	5.72[kWh]	37.76[kWh]	38.4[kWh]
N_{sb}	1,101	167	165
C_{bp}	8,368[Ah]	8,350[Ah]	8,250[Ah]
N_{bp}	64,969	9,853	3,300

Table 2.13: Performance parameters of the large LFP battery bank.

Performance parameters, LFP battery bank 25% P_{nom}			
	URB1270	PSL-SC-12500	U27-36XP
I_{sc}	15[A]	50[A]	50[A]
I_{bpc}	16.52[kA]	8.35[kA]	8.25[kA]
P_{bpc}	12,386[kW]	6,263[kW]	6,188[kW]
I_{spc}	55[A]	140[A]	150[A]
I_{bpp}	60.56[kA]	23.38[kA]	24.75[kA]
P_{bpp}	45,416[kW]	17,535[kW]	18,563[kW]
m_{bp}	64,960[kg]	62,074[kg]	61,710[kg]
vol_{bp}	59.05[m ³]	55.10[m ³]	39.08[m ³]

2.4.6 Calculated LTO battery banks

Finally, the alternatives found for LTO are analyzed using Table 2.14, which shows the technical specifications.

Table 2.14: Technical specification of LTO Batteries.

LTO Batteries			
	SLB12400L151	SLB08115L140	ELB 1500mAh LTO
V_{bc}	2.4[V]	2.4[V]	2.3[V]
E_{bc}	0.36[Wh]	0.03[Wh]	3.6[Wh]
C_{bc}	0.15[Ah]	0.014[Ah]	1.5[Ah]
$C\text{-rate}_{bcc}$	1 [h ⁻¹]	0.5 [h ⁻¹]	1 [h ⁻¹]
$C\text{-rate}_{bpc}$	20 [h ⁻¹]	20 [h ⁻¹]	8 [h ⁻¹]
m_{bc}	0.09[kg]	0.0012[kg]	0.046[kg]
vol_{bc}	$0.02 \cdot 10^{-3}$ [m ³]	$0.002 \cdot 10^{-3}$ [m ³]	$0.02 \cdot 10^{-3}$ [m ³]

a) LTO Small Battery Bank (Battery Bank 15% P_{nom})

The essential electrical characteristics, performance parameters, and physical characteristics of the small LTO battery bank are presented below in Tables 2.15, 2.16, and ??, respectively.

Table 2.15: Essential parameters of the small LTO battery bank.

Essential parameters, LTO battery bank 15% P_{nom}			
	SLB12400L151	SLB08115L140	ELB 1500mAh LTO
N_{sc}	313	313	327
E_{bs}	113[Wh]	10.52[Wh]	1.18[kWh]
N_{sb}	24,850	266,241	2,379
C_{bp}	3,727[Ah]	3,727[Ah]	3,568[Ah]
N_{bp}	7,778,050	83,333,433	777,933

Table 2.16: Performance parameters of the small LTO battery bank.

Performance parameters, LTO battery bank 15% P_{nom}			
	SLB12400L151	SLB08115L140	ELB 1500mAh LTO
I_{sc}	150[mA]	7[mA]	1.5[A]
I_{bpc}	3.73[kA]	1.86[kA]	3.57[kA]
P_{bpc}	2,795[kW]	1,398[kW]	2,676[kW]
I_{spc}	3[A]	280[mA]	12[A]
I_{bpc}	74.55[kA]	74.55[kA]	28.55[kA]
P_{bpc}	55,913[kW]	55,910[kW]	21,411[kW]
m_{bp}	700,025[kg]	100,000[kg]	35,785[kg]
vol_{bp}	152.72[m ³]	192.68[m ³]	13.15[m ³]

b) LTO Large Battery Bank (Battery Bank 25% P_{nom})

The essential electrical characteristics, performance parameters, and physical characteristics of the large LTO battery bank are presented below in Tables 2.17 and 2.18, respectively.

Table 2.17: Essential parameters of the large LTO battery bank.

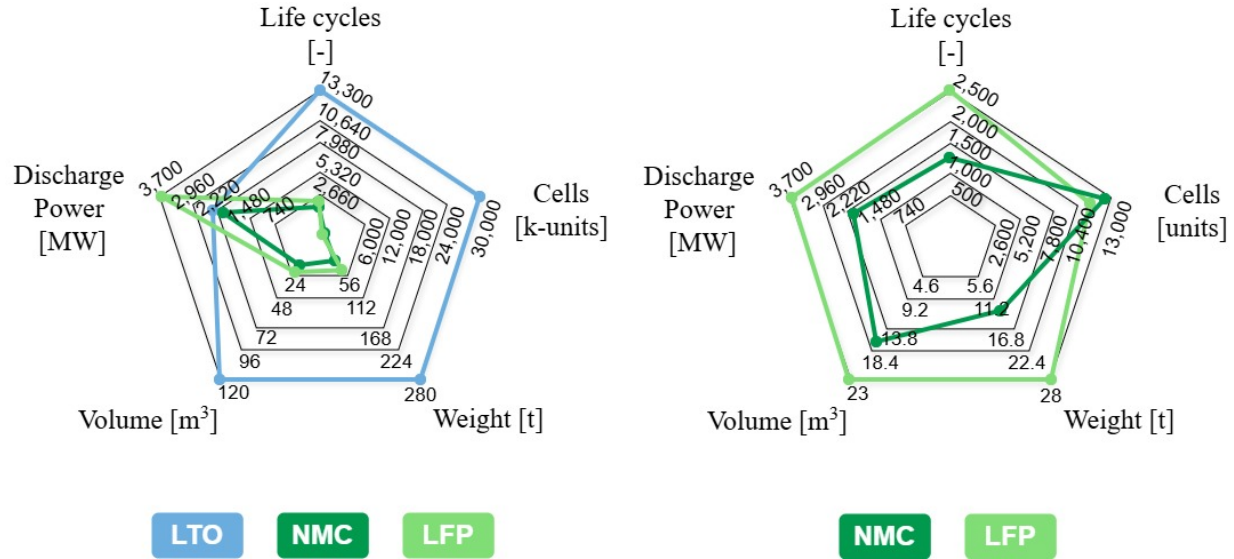
Essential parameters, LTO battery bank 25% P_{nom}			
	SLB12400L151	SLB08115L140	ELB 1500mAh LTO
N_{sc}	313	313	327
E_{bs}	113[Wh]	10.52[Wh]	1.18[kWh]
N_{sb}	55,911	599,042	5,352
C_{bp}	8,386[Ah]	8,387[Ah]	8,029[Ah]
N_{bp}	17,500,143	187,500,146	1,750,104

Table 2.18: Performance parameters of the large LTO battery bank.

Performance parameters, LTO battery bank 25% P_{nom}			
	SLB12400L151	SLB08115L140	ELB 1500mAh LTO
I_{sc}	150[mA]	7[mA]	1.5[A]
I_{bpc}	8.38[kA]	4.19[kA]	8.03[kA]
P_{bpc}	6,290[kW]	3,145[kW]	6,021[kW]
I_{spc}	3[A]	280[mA]	12[A]
I_{bpc}	167.73[kA]	167.73[kA]	64.22[kA]
P_{bpc}	125,800[kW]	125,800[kW]	48,168[kW]
m_{bp}	1,575,013[kg]	225,000[kg]	80,505[kg]
vol_{bp}	343.61[m ³]	433.54[m ³]	29.59[m ³]

2.4.7 Comparison between technologies

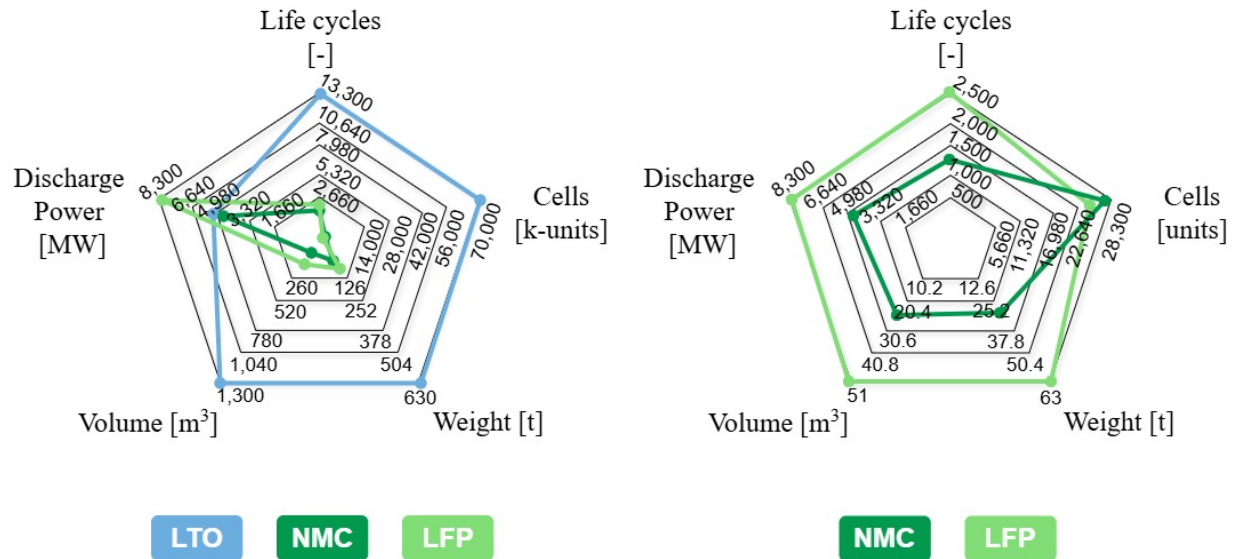
Based on the technical information provided by battery suppliers and on the calculations performed, the values obtained in each calculation for each technology are averaged to obtain average parameters for comparing technologies.



(a) Radar chart showing a technical comparison of small battery bank designs using the three technologies (average).

(b) Radar chart showing a technical comparison of small battery pack designs using NMC and LFP technologies (average).

Figure 2.1: Comparison of technologies for the technical analysis of small battery bank design.



(a) Radar chart showing a technical comparison of large battery bank designs using the three technologies (average).

(b) Radar chart showing a technical comparison of small battery pack designs using NMC and LFP technologies (average).

Figure 2.2: Comparison of technologies for the technical analysis of large battery bank design.

For comparison purposes, based on the options chosen for each technology, an average value is obtained for the technical parameters of interest to be observed, in order to provide a general analysis of the performance of each technology for the design of large and small banks.

What can be observed in Figures 2.1a and 2.2a is that LTO technology significantly outperforms

most technical parameters of interest. This is because small cells were chosen for the battery banks, which can pose greater technical challenges in their implementation.

Thus, a comparison is made only between NMC and LFP technologies through Figures 2.1b and 2.1b. In general, if technology performance is paramount, both in terms of service life and discharge power, LFP is the appropriate choice. But if there are physical restrictions within the vessel, NMC is the most suitable technology.

2.4.8 Mechanical design considerations

The mechanical design is focused on the calculation of the total weight and volume of the battery bank [8].

As mentioned in section ??, **gross tonnage (GT)** is a measure of a ship's total internal volume, while **net tonnage (NT)** is a ship's gross tonnage minus the space occupied by crew quarters, machinery, navigation equipment, engine rooms, and fuel storage. Therefore, net tonnage represents the space available for passenger, accommodation and cargo storage.

Additionally, **deadweight tonnage (DWT)** measures how much weight a vessel can safely carry, including the combined weight of cargo, fuel, fresh water, ballast water, provisions, passengers, and crew.

Thus, the mechanical parameters of the vessel are presented below in Table 2.19.

Table 2.19: Mechanical parameters of the Chilean tugboat

Deadweight [kg]	Gross Tonnage [m ³]	Net Tonnage [m ³]
220,000	761.81	228.54

It is necessary to take into account the volume and weight of the main generator presented in section ??, in order to consider the available space and the weight allowed for the installation of the batteries. These values are presented in table 2.20.

Table 2.20: Mechanical parameters of MTU 16V 4000.

Weight [kg]	Volume [m ³]
9,050	10.65

In this way, Table 2.21 shows the available space and the permitted weight for the designed battery bank.

Table 2.21: Permitted dimensions inside the vessel.

Allowed weight [kg]	Available space [m ³]
210,950	217.88

In this way, battery banks of different technologies that could be mechanically enabled for incorporation into the vessel are identified. Options that account for less than 10% of the vessel's capacity are highlighted in green (Table 2.21), while technologies that, given their technical characteristics, do not qualify as solutions are highlighted in red. This is shown in Table 2.22 and 2.23.

Table 2.22: Physical aspects of technologies (small battery banks).

Battery Bank 15% P_{nom}		
	m_{bp}	vol_{bp}
ABL-012010P	18,698[kg]	11.45[m ³]
ABL-012030P	16,405[kg]	8.80[m ³]
E4050T1	3,143[kg]	28.11[m ³]
URB1270	28,910[kg]	26.27[m ³]
PSL-SC-12500	27,877[kg]	24.73[m ³]
U27-36XP	27,302[kg]	17.28[m ³]
SLB12400L151	700,025[kg]	152.72[m ³]
SLB08115L140	100,000[kg]	192.68[m ³]
ELB 1500mAh LTO	35,785[kg]	13.15[m ³]

Table 2.23: Physical aspects of technologies (large battery banks).

Battery Bank 25% P_{nom}		
	m_{bp}	vol_{bp}
ABL-012010P	40,034[kg]	25.75[m ³]
ABL-012030P	36,779[kg]	19.73[m ³]
E4050T1	7,096[kg]	28.11[m ³]
URB1270	64,960[kg]	59.05[m ³]
PSL-SC-12500	62,074[kg]	55.10[m ³]
U27-36XP	61,710[kg]	39.08[m ³]
SLB12400L151	1,575,013[kg]	343.61[m ³]
SLB08115L140	225,000[kg]	433.54[m ³]
ELB 1500mAh LTO	80,505[kg]	29.59[m ³]

2.5 Economic Analysis

To analyze the economic viability of a hybrid tugboat, it is necessary to first understand the higher initial capital expenditures (CAPEX) and operating expenses (OPEX).

In addition, two additional factors related to emissions taxes and government subsidies are taken into account to analyze how this affects economic indicators and how sensitive they are:

- A CO₂ emissions tax of \$50/tCO₂.
- A government subsidy of 10% of CAPEX.

2.5.1 Capital Expenditure (CAPEX)

One of the major problems facing the maritime transport sector with regard to the advancement of hybrid propulsion is how to achieve a fully satisfactory implementation of a cost-effective hybrid propulsion plant [83]. The main challenge is CAPEX, i.e., keeping the capital expenditure of hybrid equipment in line with that of conventional propulsion plants.

The cost of marine battery systems is significantly higher than automotive packs due to the “marine tax”—the cost of ruggedization, certification, and low-volume manufacturing.

BESS is a mature technology and there are already hybrid tugboats that have been deployed [74]. A BESS contains Li-ion batteries, a battery management system, ventilation/cooling, and a dedicated fire extinguishing system. The BESS must be installed in a dedicated compartment with ventilation. However, Li-ion batteries have a very low energy density, and large battery installations will entail space and weight costs on the vessel. Fire safety is also a concern, as battery fires can be difficult to extinguish. The lifetime of the BESS is also limited by the number of charge/discharge cycles. The typical BESS lifetime is 8,000 to 12,000 cycles.

It is important to note that this is the estimated cost for the first tugboat [74]. This distinction is important because the first hybrid tugboat is considered a novel design, which requires additional engineering and design work hours. Therefore, 3.5% of the gross cost will be considered, which is a similar engineering cost in proportion to that seen in other projects [74].

The following are the main elements that are considered for the hybridization and retrofitting of a conventional tugboat.

a) Battery Bank NMC

Their life cycles are around 1,000–2,500 cycles. In a heavy-duty tug application with frequent peaks, NMC packs may degrade faster, necessitating a costly replacement mid-lifecycle (Year 7–10), which severely impacts the NPV [75].

- **System Price (2024/2025)** [80]: US\$600 – US\$1000 per kWh.

The higher cost is due to the use of expensive raw materials such as cobalt and the more complex manufacturing processes required to achieve the high energy density and performance characteristics [80]. For example, in electric vehicles where space and weight are critical factors, NMC batteries are often preferred despite the higher cost, as they can provide longer driving ranges without adding excessive bulk to the vehicle.

b) Battery Bank LFP

While LFP has a lower energy density (90 to 160 Wh/kg) than NMC, it offers a superior cycle life, typically achieving 2,000 to 6,000+ cycles. In many maritime applications, such as cargo vessels or mid-speed ferries, the slightly higher weight of LFP is an acceptable trade-off for the increased safety and the ability to avoid replacement for 10 years or more [76].

- **System Price (2024/2025)** [80]: US\$300 – US\$600 per kWh.

The lower cost can be attributed to the use of more abundant and less expensive raw materials, as well as a simpler manufacturing process [80]. Although they have a slightly lower energy density compared to NMC batteries, they are a popular choice for applications where safety and long-term reliability are of utmost importance, such as in home energy storage systems and some electric vehicles where cost is a significant consideration.

c) Battery Bank LTO

LTO systems can withstand 10,000 to 15,000+ cycles and are capable of ultra-fast charging, often reaching full capacity in 10 to 30 minutes [76].

- **System Price (2024/2025)** [80]: US\$800 – US\$1,200 per kWh.

The higher cost is mainly due to the production process of lithium titanate, which is more complex and requires specialized equipment. Additionally, the raw materials used in LTO batteries can be costly [80]. Despite their high cost, LTO batteries are finding applications in areas where rapid charging and high power output are crucial, such as in some electric buses and high-performance electric vehicles.

d) Power Electronics

- **DC/DC Converters:** The DC/DC converters convert and regulate the power supplied from the lithium ion batteries to the voltage requirement of the DC bus [77].
 - **System Price** [78] [79]: US\$100 – US\$400 per kW.
- **Inverters:** Bidirectional AFE that regulates the voltage from a battery-linked DC/DC converter to an AC propulsion bus, with marine-grade and liquid-cooled drives [85].
 - **System Price:** US\$250 – US\$450 per kW.
- **Variable Frequency Drives:** AC drives function as the primary interface between the electrical distribution system—whether it be a traditional AC grid or a modern common DC bus—and the three-phase motors that provide mechanical thrust [86].
 - **System Price:** US\$120 – US\$250 per kW.

e) Main genset and Electric motors

- **Main Genset:** The price ranges for diesel generators with a power output of 2000 kW (2 MW) vary significantly based on the manufacturer (Tier 1 vs. budget brands), application (Marine vs. Industrial), and condition (New vs. Used).
 - **System Price:** US\$75 – US\$750 per kW.
- **Electric Motors:** The price for a 1,000 kW (1 MW) squirrel cage induction motor suitable for hybrid tugboat propulsion varies significantly depending on the manufacturer (ABB, Siemens (SIMOTICS M), WEG, Danfoss Editron), cooling method, and classification status.
 - **System Price:** US\$90 – US\$250 per kW.

f) Integration and Safety systems

- **Integration/cabling:** It is considered to be around 20% of the cost of equipment.
 - **System Price:** US\$100,000 – US\$175,000.

- **Fire/Safety Systems:** Battery rooms must be A-60 fire-rated, gas-tight, and equipped with explosion relief vents and specialized ventilation to handle potential off-gassing, adding structural steel and HVAC costs [87].

– **System Price:** US\$50,000 – US\$80,000.

g) Total Cost

Tables 2.24 and 2.25 show the total cost obtained for the different elements identified for both the small battery bank and the large battery bank, using an average of the highest and lowest costs for each element.

Table 2.24: BESS implementation costs, without taxes or subsidies (small bank).

Battery Bank 15% P_{nom}	
NMC Battery Bank	US \$2,240,000
LFP Battery Bank	US \$1,260,000
LTO Battery Bank	US \$2,800,000
DC/DC Converters	US \$150,000
Inverter and Drives	US \$580,000
Main Genset	US \$825,000
Electric Motors	US \$340,000
Integration/Cabling/Equipment	US \$137,500
Fire/Safety Systems	US \$65,000

Table 2.25: BESS implementation costs, without taxes or subsidies (small bank).

Battery Bank 25% P_{nom}	
NMC Battery Bank	US \$5,040,000
LFP Battery Bank	US \$2,835,000
LTO Battery Bank	US \$6,300,000
DC/DC Converters	US \$250,000
Inverter and Drives	US \$720,000
Main Genset	US \$825,000
Electric Motors	US \$340,000
Integration/Cabling/Equipment	US \$137,500
Fire/Safety Systems	US \$65,000

Using the cost information provided in the tables, the cost associated with engineering is calculated, considered to be 3.5% of the total costs. This can be seen in tables 2.26 and 2.27.

Table 2.26: Engineering cost associated with the small battery bank.

Battery Bank 15% P_{nom}	
Engineering (NMC)	US \$151,812
Engineering (LFP)	US \$117,512
Engineering (LTO)	US \$171,412

Table 2.27: Engineering cost associated with the large battery bank.

Battery Bank 25% P_{nom}	
Engineering (NMC)	US \$258,212
Engineering (LFP)	US \$181,037
Engineering (LTO)	US \$302,312

Engineering costs will be added to the costs obtained for hybrid propulsion system equipment, thus obtaining the CAPEX for each battery bank. This can be seen in Tables 2.28 and 2.29.

Table 2.28: Calculated CAPEX, without taxes or subsidies (small bank).

Battery Bank 15% P_{nom}	
CAPEX _{NMC}	US \$4,489,313
CAPEX _{LFP}	US \$3,239,988
CAPEX _{LTO}	US \$4,726,088

Table 2.29: Calculated CAPEX, without taxes or subsidies (large bank).

Battery Bank 25% P_{nom}	
CAPEX _{NMC}	US \$7,635,713
CAPEX _{LFP}	US \$5,353,538
CAPEX _{LTO}	US \$8,939,813

This additional capital expenditure must be recouped through reduced operating costs resulting from the use of electricity instead of diesel fuel and reduced maintenance [84]. The local price of electricity and the local price of fuel are very important for the business case. This improves substantially when authorities provide cheap green electricity at the port to achieve their zero-emission targets.

2.5.2 Operational Expenditure (OPEX)

OPEX savings are derived primarily from fuel reduction during "loitering" (low load) and maintenance intervals extended by reduced engine runtime [83].

Thus, the following variables are considered in operating expenses.

a) Fuel Consumption Savings

A conventional tugboat consumes approximately several thousand liters of fuel just idling during its daily workday. A hybrid tugboat in electric mode consumes 0 liters in that mode. The batteries allow the load on the main engines to be balanced, thereby improving the specific fuel consumption (BSFC).

b) Maintenance Engine Reduction

Hybrid tugs can reduce main engine operating hours by up to 60% per year by using batteries for transit and standby mode. This extends the overhaul intervals for propulsion engines, reducing

annualized maintenance costs. However, these savings in propulsion engine maintenance are partially offset by the need to budget for battery replacement, so an “amortization fund” must be established in the financial model to cover battery pack replacement [84]. For NMC/LFP, replacement will be considered after 15 years, while for LTO, a useful life of 30 years will be considered, exceeding the project period, so no replacement cost will be amortized for this technology.

c) Total Costs

To calculate operating costs, a fuel cost of \$0.8/L will be used, and the annual consumption for both one maneuver and two maneuvers will be averaged. These consumption figures are obtained from section ??.

Table 2.30: Operating costs, excluding taxes and subsidies (small bank).

Battery Bank 15% P_{nom}		
	Conventional Tug	Hybrid Tug
Total fuel cost	US \$1,715,646	US \$974,404
Electricity	-	US \$30,000
Engine Maintenance	US \$40,000	US \$20,000
NMC Battery Replacement	-	US \$149,400
LFP Battery Replacement	-	US \$84,000
LTO Battery Replacement	-	-

Table 2.31: Operating costs, excluding taxes and subsidies (large bank).

Battery Bank 25% P_{nom}		
	Conventional Tug	Hybrid Tug
Total fuel cost	US \$1,715,646	US \$870,014
Electricity	-	US \$30,000
Engine Maintenance	US \$40,000	US \$20,000
NMC Battery Replacement	-	US \$336,000
LFP Battery Replacement	-	US \$189,000
LTO Battery Replacement	-	-

With the operating costs, we proceed to calculate the OPEX for each technology, which can be seen in Tables 2.32 and 2.33.

Table 2.32: Calculated OPEX (annual), without taxes or subsidies (small bank).

Battery Bank 15% P_{nom}	
OPEX _{NMC}	US \$1,173,737
OPEX _{LFP}	US \$1,108,404
OPEX _{LTO}	US \$1,024,404

Table 2.33: Calculated OPEX (annual), without taxes or subsidies (large bank).

Battery Bank 25% P _{nom}	
OPEX _{NMC}	US \$1,256,014
OPEX _{LFP}	US \$1,109,014
OPEX _{LTO}	US \$920,014

Finally, the annual benefit can be obtained by calculating the difference between the OPEX of the conventional tugboat (OPEX_{tug}) and the OPEX of each technology. These results are shown in Tables 2.34 and 2.35.

Table 2.34: Calculated profit (annual), without taxes or subsidies (small bank).

Battery Bank 15% P _{nom}	
OPEX _{TUG}	US \$1,755,646
Profit (NMC)	US \$581,909
Profit (LFP)	US \$647,242
Profit (LTO)	US \$731,242

Table 2.35: Calculated profit (annual), without taxes or subsidies (large bank).

Battery Bank 25% P _{nom}	
OPEX _{TUG}	US \$1,755,646
Profit (NMC)	US \$499,632
Profit (LFP)	US \$646,632
Profit (LTO)	US \$835,632

The economic benefit provides insight into economic sensitivity indicators, which are essential for analyzing the life of a project.

2.5.3 Financial Performance

Based on the information described above, the viability of the project can be determined using economic indicators such as net present value (NPV), internal rate of return (IRR), and finally return on investment (ROI).

a) Net Present Value (NPV)

The following equation will be used to calculate the NPV.

$$NPV = \sum_{t=1}^{LFS} \frac{Prf}{(1 + ds)^t} - Inv. \quad (2.20)$$

Where:

- **LFS:** Lifespan of the project (years).
- **Prf:** Annual profit.

- **ds**: Discount rate.
- **Inv**: Initial Investment.

For all scenarios, a **discount rate** of 8% and a useful life of 10 and 20 years for the hybrid tugboat will be proposed. In this way, the positive NPV will demonstrate that the project will generate a return on investment.

In addition, it is proposed that the batteries used will have a lifespan of 3,000 to 5,000 cycles during the lifetime of this project.

This provides the net present values for each technology and for each battery bank requested.

Table 2.36: Calculated NPV, without taxes or subsidies (small bank).

Battery Bank 15% P _{nom}			
Lifespan	10 years	15 years	20 years
NPV _{NMC}	−\$584,658	\$491,522	\$1,223,953
NPV _{LFP}	\$1,103,059	\$2,300,067	\$3,114,730
NPV _{LTO}	\$180,606	\$1,532,963	\$2,453,354

Table 2.37: Calculated NPV, without taxes or subsidies (large bank).

Battery Bank 25% P _{nom}			
Lifespan	10 years	15 years	20 years
NPV _{NMC}	−\$4,283,141	−\$3,359,123	−\$2,730,252
NPV _{LFP}	−\$1,014,584	\$181,295	\$995,191
NPV _{LTO}	−\$2,766,444	−\$1,064,972	\$93,414

Tables 2.36 and 2.37 show that for a small battery bank, the project is economically viable for all technologies over a short period of time. This contrasts with the large battery bank, where only LFP technology is viable, NMC is viable over a period of 15 years, and LTO is not viable in any of the proposed periods.

The case is also analyzed in terms of how it is affected by emissions taxes and government grants, effects that can be applied jointly or separately. This information is provided in Appendix ??, Appendix ??, and Appendix ??.

b) Internal Rate of Return (IRR)

IRR is an indicator of a project's profitability that is used to decide whether to accept or reject an investment project [88]. To do this, the IRR is compared to a minimum rate or discount rate. If the project's rate of return—expressed by the IRR—exceeds the discount rate, it is accepted; otherwise, it is rejected. The IRR can be calculated using the following equation:

$$0 = \sum_{t=1}^{LFS} \frac{Prf}{(1 + IRR)^t} - Inv. \quad (2.21)$$

Where:

- **LFS**: Lifespan of the project (years).

- **Prf**: Annual profit.
- **Inv**: Initial Investment.
- **IRR**: Internal rate of return.

The results obtained for this economic variable for each design are shown below.

Table 2.38: Calculated IRR, without taxes or subsidies (small bank).

Battery Bank 15% P _{nom}			
Lifespan	10 years	15 years	20 years
IRR _{NMC}	5.02%	9.75%	11.49%
IRR _{LFP}	15.07%	18.39%	19.40%
IRR _{LTO}	8.84%	13.00%	14.43%

Table 2.39: Calculated IRR, without taxes or subsidies (large bank).

Battery Bank 25% P _{nom}			
Lifespan	10 years	15 years	20 years
IRR _{NMC}	-7.05%	-0.23%	2.71%
IRR _{LFP}	3.59%	8.55%	10.41%
IRR _{LTO}	-1.21%	4.56%	6.87%

For the cases presented in Tables 2.38 and 2.39, it can be seen that all small battery banks manage to exceed the discount rate imposed by the investment, even though LTO technology slightly exceeds the discount rate over a 10-year period. In the case of large battery banks, only LFP manages to be profitable within the 10-year period, and extending the period allows for the viability of NMC and LFP projects, which is not the case with LTO technology, even when considering a 20-year period.

The case is also analyzed in terms of how it is affected by emissions taxes and government grants, effects that can be applied jointly or separately. This information is provided in Appendix ??, Appendix ??, and Appendix ??.

c) Return on Investment (ROI)/ Payback Period

Return on investment (ROI) is a measure of profitability. It indicates how much profit or loss a company makes from an investment in relation to its cost. Measuring return on investment helps identify which projects truly drive profitability, optimize resources, and strengthen the company's competitiveness.

The formula is as follows:

$$ROI = \frac{G - Inv}{Inv} \cdot 100 \quad (2.22)$$

- **G**: Annual gain from investment.
- **Inv**: Initial Investment.

To calculate the period in which the investment is recouped and profits are generated P_{ROI} , the following formula is used:

$$P_{ROI} = \frac{Inv}{G} \quad (2.23)$$

This provides the periods for each technology in which the return on investment begins to be positive. This can be seen in Tables X and Y.

Table 2.40: Calculated period for positive ROI, without taxes or subsidies (small bank).

Battery Bank 15% P_{nom}	
	Period
$ROI_{NMC} > 0$	≈ 7 years and 9 months
$ROI_{LFP} > 0$	≈ 5 years
$ROI_{LTO} > 0$	≈ 6 years and 6 months

Table 2.41: Calculated period for positive ROI, without taxes or subsidies (large bank).

Battery Bank 25% P_{nom}	
	Period
$ROI_{NMC} > 0$	≈ 14 years and 9 months
$ROI_{LFP} > 0$	≈ 8 years
$ROI_{LTO} > 0$	≈ 10 years and 4 months

Based on the ROI financial indicator, it can be seen that LFP technology is the one that allows profits to be obtained in a shorter period of time, both for small battery banks and large battery banks. An increase in fuel prices allows for higher annual profits when comparing conventional systems with hybrid systems, which in turn also shortens the payback period. Conversely, high onshore electricity rates can prolong the payback period.

The case is also analyzed in terms of how it is affected by emissions taxes and government grants, effects that can be applied jointly or separately. This information is provided in Appendix ??, Appendix ??, and Appendix ??.

2.6 Chapter Summary

This chapter has the focus on the preliminary design, technical evaluation, and economic viability of an Energy Storage System (ESS) for a hybrid tugboat. The chapter outlines the methodology for sizing battery banks and compares three primary lithium-ion technologies: NMC, LFP, and LTO.

1. **Sizing Methodology and Requirements:** The design process begins by calculating the required energy capacity based on the tugboat's operational load profile (E_{load}), which includes both propulsion and "hotel" (auxiliary) loads. To determine the final installed capacity ($E_{installed}$), it was applied several correction factors.
2. **Technology Comparison:** It was compared Nickel Manganese Cobalt (NMC), Lithium Iron Phosphate (LFP), and Lithium Titanate Oxide (LTO), based on market options. The comparison of alternatives was carried out in a particularly technical manner.

3. **Economic and Financial Results:** The economic analysis considers the Capital Expenditure (CAPEX) of retrofitting a tugboat and the Operational Expenditure (OPEX) savings from reduced fuel consumption.

The conclusion of the design analysis is that LFP technology is the most viable solution for hybrid tugboats, as it balances lower initial cost with a long service life, enabling a faster return on investment compared to NMC or LTO.

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